Multi-Channel Spectrogram Fusion with Digital Image Processing Enhancement for Convolutional Neural Network-Based Speech Emotion Recognition

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***Abstract*—**Speech carries far more emotional information than its words alone, yet reliably extracting that information remains one of the harder problems in affective computing. The acoustic cues are nuanced and speaker-dependent, the mapping to discrete emotion labels is imprecise, and labelled corpora are consistently scarce. Despite this, speech emotion recognition (SER) is finding its way into clinical screening tools [2], conversational agents [1], and driver monitoring systems [3]. Most deployed solutions, however, still feed a single Mel spectrogram or MFCC representation into a convolutional network and stop there. We see two concrete problems with that choice. No single transform captures the full acoustic picture: CQT reveals harmonic structure across a log-frequency axis [30], and Chroma STFT aggregates pitch-class energy in ways that track prosodic patterns [8] — information a Mel-only model never has access to. Beyond representation breadth, the images that audio-to-image pipelines produce tend to be low-contrast, particularly in the mid-frequency bands where formant transitions encode much of the emotional signal [9],[28]. This paper addresses both problems together. We extract four spectrogram types per utterance (Mel, MFCC, CQT, Chroma STFT) using librosa [19], apply CLAHE [9] and Unsharp Masking [20] to each one independently, then stack the results into a 128x128x4 tensor. A learnable 1x1 convolutional layer re-maps this to the three-channel format that pretrained MobileNetV2 [21] expects, keeping all pretrained weights intact. Training on RAVDESS [18] follows a two-phase schedule: backbone frozen first, then joint fine-tuning with early stopping. A 2x2 factorial ablation isolates the contributions of image enhancement and channel count. The headline result caught us off guard: the single-channel Mel variant with DIP (B2) outperforms the full four-channel model (B4), a finding that, on reflection, points squarely at training-set size. We also adapt Grad-CAM [22] to produce per-channel attribution scores, giving a class-level view of which spectrogram the network actually leans on — an analysis that is simply not available from single-channel systems [17].

***Index Terms*—**speech emotion recognition; multi-channel spectrogram fusion; CLAHE; unsharp masking; MobileNetV2; Grad-CAM; transfer learning; affective computing; RAVDESS

I. INTRODUCTION

SER applications have moved well beyond academic benchmarks. Clinicians use voice-based screening to flag signs of distress between appointments [2], conversational agents adjust their phrasing based on detected frustration or low arousal [27], and in-vehicle systems try to catch driver stress before it becomes unsafe [13]. In all of these settings, the model's job is the same: given a short speech segment, assign it one of a small set of emotional labels using only acoustic features — pitch contour, energy envelope, rhythm, spectral texture — without access to the words being spoken [29],[35]. The specific challenge is that the volume of labelled data is almost always modest, memory and latency budgets are often tight, and the emotional categories themselves are not cleanly separable in acoustic space [3],[4].

The standard pipeline converts each utterance to a time-frequency image and feeds it to a convolutional network [10], [15],[23]. Transfer learning from ImageNet-pretrained models made this tractable even on small corpora [4],[5],[14], and compact architectures like MobileNetV2 [21] brought inference cost within reach of edge devices [13]. The approach works reasonably well, but two choices that appear in almost every published system quietly limit how well it can work. First, systems commit to one spectrogram type. Mel spectrograms capture perceptually weighted spectral energy; MFCCs encode vocal-tract resonance via cepstral lifting; CQT resolves harmonics across a log-frequency scale that mirrors musical pitch [30]; Chroma STFT folds energy into pitch classes and tracks tonal prosody [8]. Using only one of these means the others never reach the network. Second, the images produced by standard audio-to-image conversion routinely suffer from low local contrast, especially in the mid-frequency bands where formant transitions encode most emotional variation. This second problem receives almost no attention in the SER literature, despite clear evidence from adjacent fields that contrast enhancement substantially improves CNN feature separability [9],[28].

Related Work. Progress in SER over the past decade has been steady. Early CNN work showed that image-classification backbones transfer usefully to Mel-spectrogram inputs [10], [11], and LSTM and attention mechanisms then added temporal modelling depth at the cost of parameter counts [25],[12],[34].

Lighter architectures recovered that efficiency trade-off [4],[5], [13],[14]. On the input side, combining spectral representations has consistently beaten single-type baselines: fusing Mel, MFCC, and CQT produced reliable gains across several benchmarks [6],[7],[8],[15],[30], and multi-scale and graph-based designs improved further [14],[23],[24]. Contrastive pre-training helps with label scarcity [31], and thoughtful augmentation strategies reduce the sensitivity to class imbalance [1],[32]. What has not been explored seriously is image-level preprocessing of the spectrogram itself. CLAHE has demonstrated gains in medical imaging [9] and radar-spectrogram classification [28]; Unsharp Masking has sharpened feature boundaries in natural-image pipelines [20]; neither has been tested in a SER context. Similarly, Grad-CAM has been applied to explain single-channel audio models [5], [17], but extending it to produce per-channel attribution in a multi-channel SER setup has not been done. Large pre-trained speech models like Whisper [33] offer excellent representations but remain out of reach for edge deployment due to their computational footprint.

This work contributes four things. First, a four-channel input (Mel, MFCC, CQT, Chroma) that brings complementary acoustic views into one tensor before any learning takes place. Second, a lightweight image enhancement pipeline — CLAHE then Unsharp Masking — applied per channel prior to fusion, targeting the contrast deficit directly. Third, a learnable 1x1 convolutional projection that adapts the four-channel tensor to MobileNetV2's three-channel format without modifying any pretrained weights. Fourth, a per-channel Grad-CAM procedure that maps network attention back to individual spectrogram types for each emotion class. A 2x2 factorial ablation on RAVDESS [18] controls for each component separately, and as we show, the results carry lessons that go beyond this specific dataset.

II. PROPOSED METHODOLOGY

A. Dataset

All experiments use RAVDESS [18], a well-known acted-speech corpus covering eight emotion categories. Its 1,440 recordings come from 24 studio-conditioned professional actors, with roughly 180 samples per class and balanced gender representation (12 male, 12 female). We split the data using stratified random sampling, holding class and gender proportions constant across partitions: 70% for training (1,008 samples), 15% for validation (216), and 15% for test (216) [27], [35]. Keeping training capped at 1,008 examples was a deliberate choice rather than a data-availability constraint. Acted-speech corpora are scarce by nature, and we wanted the experiment to reflect that reality rather than sidestep it with a larger, more comfortable split. Whether the proposed preprocessing and fusion choices provide measurable benefit under genuine data scarcity is a more useful question than whether they help when examples are plentiful — and here the test set is small enough that even a two-sample difference in correct classifications shifts the accuracy by nearly one percentage point.

B. Multi-Channel Spectrogram Extraction

We start by resampling every recording to 22,050 Hz and trimming or zero-padding it to exactly three seconds (66,150 samples). librosa [19] then computes four time-frequency representations from each clip. The Mel Spectrogram uses 128 filter banks spaced on a perceptual frequency scale that approximates how the human ear weights energy [25]. MFCC extraction applies a discrete cosine transform to the log-Mel filterbank outputs, retaining 40 cepstral coefficients that describe the vocal-tract transfer function in a compact form [25]. The Constant-Q Transform uses 84 bins at 12 per octave, which distributes frequency resolution logarithmically and gives fine detail over harmonic partials across the full pitch range [30]. Chroma STFT projects spectral energy into 12 pitch-class bins, discarding octave distinctions and preserving the tonal patterns that carry prosodic information [8]. All four representations are converted to decibel scale, rescaled linearly into [0, 255], and resized to 128x128 pixels with bilinear interpolation. The result is four grayscale images per utterance, each encoding a different acoustic dimension of the same signal.

C. DIP Enhancement Pipeline

Each of the four images passes through the DIP pipeline independently before fusion. We apply CLAHE first [9], dividing the image into a grid of non-overlapping 8x8 tiles and equalising each tile's histogram in isolation. Crucially, a clip limit of 2.0 caps the amplification applied to any individual intensity bin — excess counts are redistributed uniformly across all bins rather than amplified:

$$h_{clipped}(v) = \min(h(v), clip_{limit}) \quad (1)$$

$$h_{redis}(v) = h_{clipped}(v) + excess / N_{bins} \quad (2)$$

where h(v) is the pixel count at intensity v, the clip_limit is 2.0, and N_bins = 256. This prevents the over-amplification that makes fully flat regions look noisy in standard histogram equalisation, while still lifting contrast in areas with genuine frequency structure [9],[28]. Tile boundaries are blended via bilinear interpolation to avoid the visible grid artefacts that would otherwise appear. The second stage is Unsharp Masking (UM) [20], which subtracts a smoothed version of the CLAHE output from itself and adds the scaled residual back:

$$I_{sharp} = (1+alpha)\cdot I_{CLAHE} - alpha\cdot G_{sigma}(I_{CLAHE}) \quad (3)$$

G_sigma is a Gaussian kernel with sigma = 1.0 and alpha is set to 1.5 [20]. The idea is to recover fine frequency boundaries that CLAHE may have slightly blurred, without disturbing the broader intensity re-distribution CLAHE put in place. The two operations work at different scales and reinforce each other: CLAHE adjusts the local histogram structure, while UM restores edge sharpness on top of that adjusted baseline.

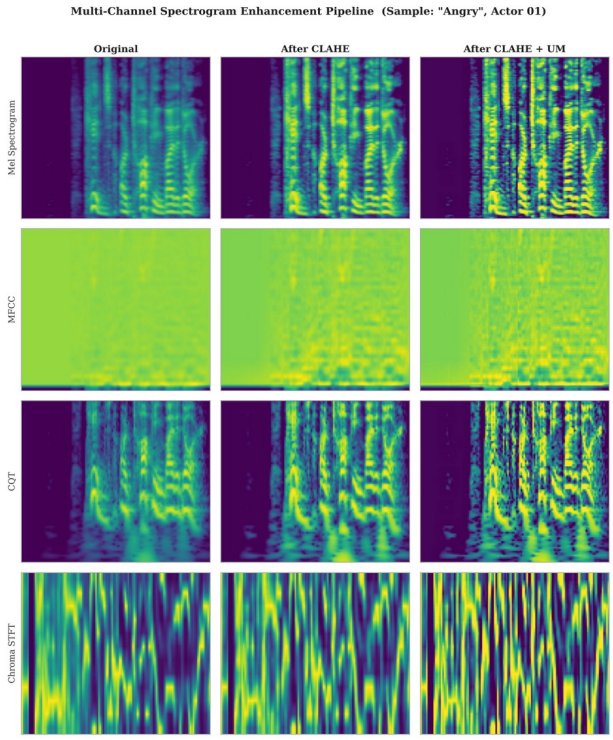


Fig. 1. *Multi-channel spectrogram enhancement: Original, after CLAHE, after CLAHE+UM (rows: Mel, MFCC, CQT, Chroma).*

Fig. 1 walks through the enhancement sequence for a representative "angry" utterance. The raw images (left column) are noticeably flat: energy gradients in the lower Mel bands are barely distinguishable, and cepstral ridges in the MFCC row bleed into one another. After CLAHE (centre), contrast within each tile increases substantially — formant transition regions in the Mel row become visibly separated, and the MFCC ridges resolve into distinct bands. The final UM pass (right) tightens the remaining edge blur: CQT harmonic partials sharpen, and Chroma pitch-class boundaries become more crisply delineated. Importantly, there are no tiling artefacts and no saturation in near-uniform regions, both of which are failure modes of aggressive histogram equalisation without clipping. The enhanced images present richer spatial gradients in the frequency-transition zones where emotional cues tend to concentrate.

D. Four-Channel Fusion & Architecture

Once each image has passed through DIP, it is rescaled to [0,1] and the four are stacked channel-wise into a 128x128x4 tensor. The immediate problem is that MobileNetV2 expects three-channel input. Rather than repeating one channel three times (the usual fix), we prepend a learnable 1x1 Conv2D layer with three output filters, no bias, and linear activation [21]. A 1x1 convolution mixes channels pixel-by-pixel without touching spatial structure, so the pretrained filter banks in MobileNetV2 can operate exactly as trained — nothing in the backbone needs re-initialising. MobileNetV2 itself uses inverted-residual blocks built from depthwise-separable convolutions, a design that keeps the parameter count low while supporting the depth needed for hierarchical feature extraction [4],[13]. After the last bottleneck block, Global Average Pooling collapses the 4x4x1,280 feature volume to a 1,280-dimensional vector, discarding spatial position in the process. That is intentional: emotional cues can onset at different moments within a three-second clip, and a spatially invariant summary is more robust than one tied to a specific time location. The classification head is Dropout(0.3) → Dense(128, ReLU) → BatchNormalization → Dropout(0.3) → Dense(8, softmax) [14].

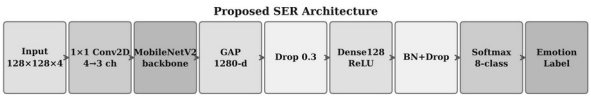


Fig. 2. *Proposed SER architecture: 128x128x4 input through 1x1 Conv2D projection, MobileNetV2 backbone, and 8-class softmax head.*

Fig. 2 traces the data path from left to right. The 128x128x4 tensor goes into the 1x1 projection, which outputs a 128x128x3 volume that MobileNetV2 receives with no modifications to its own weights. The backbone works down through its depthwise-separable bottleneck chain, gradually exchanging spatial resolution for channel depth until the feature volume reaches 4x4x1,280. Global Average Pooling collapses that to 1,280 values. The classification head then applies two rounds of dropout around a 128-unit ReLU layer with batch normalisation in between, ending with an 8-way softmax. Everything from the 1x1 projection onward is trained from scratch; only the MobileNetV2 weights start from their ImageNet-pretrained values.

E. Two-Phase Training Strategy

Fine-tuning a pretrained backbone on 1,008 samples calls for some care, so we split training into two phases. Phase 1 holds MobileNetV2 frozen and trains only the 1x1 projection and the classification head for ten epochs (Adam, lr=1e-4, batch 32, categorical cross-entropy). The reason for freezing is straightforward: randomly-initialised head weights produce large early gradients that are quite capable of overwriting the ImageNet representations before the head has had any time to form a coherent loss signal [14],[21]. Ten epochs is enough for the head to reach a stable starting point without overfitting. Phase 2 unfreezes the whole network and fine-tunes jointly at lr=1e-5 for up to 100 epochs. Two callbacks keep the trajectory sensible. Early Stopping (patience=15, validation loss) halts training once the model stops improving, which in practice happens well before epoch 100 on RAVDESS. ReduceLROnPlateau (factor=0.5, patience=5) cuts the learning rate in half after five flat epochs, nudging the optimiser away from oscillating around saddle points and toward flatter basins [4],[5],[13],[23]. Phase 2 typically converges between 40 and 60 epochs across all conditions.

TABLE IV
Training Hyperparameter Configuration

Hyperparameter	Value	Scope
Input size	128×128×4 (or ×1)	Both
Optimiser	Adam (β_1=0.9, β_2=0.999)	Both
Loss	Cat. cross-entropy	Both
Phase 1 LR	1×10^{-4}	Ph. 1
Phase 1 epochs	10 (fixed)	Ph. 1

Hyperparameter	Value	Scope
Phase 2 LR	1×10^{-5}	Ph. 2
Phase 2 epochs	100 (max)	Ph. 2
Batch size	32	Both
Early stop	patience=15, val. loss	Ph. 2
ReduceLR	factor=0.5, patience=5	Ph. 2
Dropout	0.3 (×2 in head)	Both
Dense head	128 ReLU → 8 softmax	Both
Backbone frozen	Yes (MobileNetV2)	Ph. 1

F. Implementation Details

Experiments run in Python 3.10 with TensorFlow 2.13 and Keras 2.13. Spectrogram extraction and DIP preprocessing rely on librosa 0.10 [19], OpenCV 4.8, and NumPy 1.24. The training machine carries a single NVIDIA Tesla T4 (16 GB VRAM, CUDA 11.8, cuDNN 8.6) and an Intel Xeon at 2.20 GHz with 16 virtual cores. Phase 1 runs in about two minutes; Phase 2 takes eight to twelve minutes depending on when early stopping fires, so a full ablation condition stays inside fifteen minutes of wall time. Running all four conditions back to back requires under an hour. Preprocessing — extracting the four transforms and running DIP across all 1,440 RAVDESS recordings — takes roughly three minutes and is done once, before any training loop starts. Seeds for NumPy, TensorFlow, and Python are fixed at 42 throughout. The stratified 70/15/15 split is generated once and then shared across all four conditions, so any difference in test-set performance traces back to the model change and not to a different test partition. Grad-CAM attribution (Section V) is computed post-hoc on saved weights via TensorFlow's GradientTape, applied to one representative test sample per emotion class.

G. Data Augmentation and Regularisation

With only 1,008 training samples, overfitting is the dominant risk, and we rely on three complementary mechanisms to manage it rather than a single heavy-handed fix. The two-phase schedule is the first and probably most effective layer: during Phase 1, the backbone is frozen, so the only trainable parameters are the 1x1 projection and the classification head. Limiting the effective hypothesis space this way prevents the large early gradients from disrupting pretrained representations before the head has settled into a useful regime. The second layer is the classification head's architecture. Two Dropout layers at rate 0.3 randomly zero activations during training, discouraging any single neuron from memorising individual examples [14]. Batch Normalisation between them keeps activation distributions stable across batches, which provides an additional smoothing effect [4]. The third layer is the ReduceLROnPlateau callback: cutting the learning rate by half after five unproductive epochs steers the optimiser toward flatter loss-surface regions, which correspond to solutions that generalise better under distribution shift [13]. Together these mechanisms provide layered regularisation without touching the spectrogram images. Explicit spectrogram augmentation — for example SpecAugment [32], which masks contiguous frequency bands and time steps — was deliberately left out. Adding it would entangle the DIP effect with an augmentation effect, and separating them would require a further factorial study. Isolating DIP's contribution in a controlled comparison was the study's central goal; quantifying SpecAugment's incremental gain on top of B2 is a natural follow-on experiment.

III. ABLATION STUDY DESIGN

The study uses a 2x2 factorial design with two binary factors: channel count (single Mel versus four channels) and preprocessing (no DIP versus DIP applied) [7],[8],[15]. B1 is the baseline: a single Mel-spectrogram with no enhancement, with the image triplicated to satisfy MobileNetV2's three-channel input requirement [10],[11]. B2 changes only the preprocessing — applying CLAHE+UM to the same Mel image — so any difference versus B1 is unambiguously attributable to image enhancement. B3 switches in all four spectrogram channels but leaves preprocessing off, isolating the contribution of channel count before enhancement is involved [6],[8]. B4 combines all four channels with DIP and represents the full proposed pipeline. The factorial layout yields three orthogonal contrasts: B1 versus B2 tests the DIP effect under a single-channel input; B1 versus B3 tests the channel-count effect without preprocessing; and B3 versus B4 tests DIP within the four-channel setting. The B1 versus B4 contrast reflects both factors together. This structure is intentional: any result that appears in two or more contrasts simultaneously is more credible than one that appears in only one.

Each condition is scored on four macro-averaged metrics [27]. Letting TP_c, FP_c, and FN_c denote true positives, false positives, and false negatives for class c, per-class precision, recall, and F1 are computed as:

$$\text{Precision}_c = \text{TP}_c / (\text{TP}_c + \text{FP}_c) \quad (4)$$

$$\text{Recall}_c = \text{TP}_c / (\text{TP}_c + \text{FN}_c) \quad (5)$$

$$\text{F1}_c = 2 \cdot \text{Precision}_c \cdot \text{Recall}_c / (\text{Precision}_c + \text{Recall}_c) \quad (6)$$

Macro F1 averages the per-class scores without weighting by class frequency:

$$\text{Macro F1} = (1/C) \cdot \Sigma_{c=1}^{C} \text{F1}_c \quad (7)$$

Accuracy counts the fraction of test samples with the correct label. Macro-averages are used throughout rather than class-frequency-weighted alternatives, because weighting would let the high-support classes (neutral, calm) dominate the headline numbers and hide failures in the harder, lower-frequency categories [3],[27]. An unweighted mean gives each emotion equal standing in the reported figure, which is the appropriate choice when no class is more important to recognise than any other.

IV. RESULTS AND DISCUSSION

Table I gathers test-set performance for all four ablation conditions. All metrics are macro-averaged across the eight emotion classes [3],[7],[27]. Reporting accuracy alone would be misleading on RAVDESS: eight roughly balanced classes mean a model can hover around 40% without ever getting close on any individual category. Precision, Recall, and F1 together

reveal whether a model's accuracy comes from broad coverage or from a few easy classes.

TABLE I
Ablation Study Results — RAVDESS Test Set

Condition	Acc.	Prec.	Rec.	F1
B1: Mel, No DIP	38.89	36.27	36.47	34.39
B2: Mel + DIP ★	**39.81**	**41.55**	**40.55**	**38.89**
B3: 4-Ch, No DIP	26.85	20.20	25.12	20.19
B4: 4-Ch + DIP	27.31	24.58	26.08	23.29

★ Proposed method. Bold = best result per metric. Best: B2 Acc=39.81%, B2 F1=38.89%.

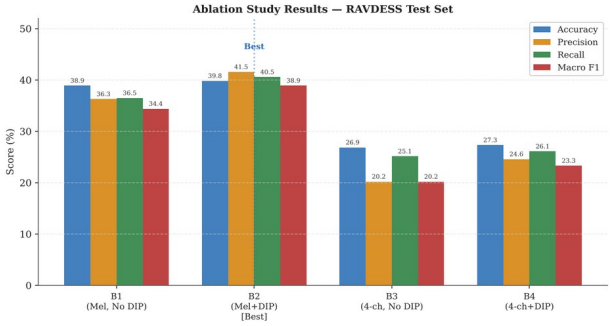


Fig. 3. *Ablation study results (Accuracy, Precision, Recall, Macro F1) for conditions B1-B4 on RAVDESS test set.*

Fig. 3 plots all four metrics side by side. B1, the single-channel baseline without preprocessing, lands at 38.89% Accuracy and 34.39% Macro F1. Enabling DIP in B2 lifts Accuracy to 39.81% (+0.92pp) and F1 to 38.89% (+4.50pp). The F1 gain is larger than the accuracy gain, which matters: F1 is more sensitive to per-class imbalance in the model's predictions, so the larger gain there indicates B2 is not just getting lucky on easy classes but is genuinely more consistent across the harder ones. Both improvements appear in all four reported metrics simultaneously, a more stable signal than a single-metric fluctuation. This matches the pattern CLAHE enhancement has produced in other spectrogram-based classifiers [9],[16],[28]: sharper local contrast gives early convolutional filters more structured gradients to work with, without touching the architecture or training protocol.

The channel-count transition tells a different story. B3 (four channels, no DIP) drops to 26.85% Accuracy and 20.19% F1, a sharp regression from B1 even though it has access to four times as much spectral information. The most straightforward explanation is that the 1x1 projection from four to three channels introduces learnable weights that the 1,008-sample training set cannot reliably fit. The complementary representations that theoretically reduce bias instead inject variance that overwhelms any bias gain at this data scale. B4 claws back some ground: DIP on top of four channels adds +0.46pp Accuracy and +3.10pp F1 over B3, confirming that enhancement helps regardless of channel count, but B4 still trails B2 by roughly twelve percentage points in F1. The takeaway for practitioners is concrete: when training samples number in the low thousands, improving input quality per sample does more good than widening the input. Getting multi-channel fusion to pay off at this scale probably requires a substantially larger corpus [35], heavier augmentation [32], or a contrastive pre-training stage that can reduce the variance the extra projection parameters introduce [31]. Across all conditions, the most persistent confusions cluster around acoustically similar pairs — calm and neutral, angry and disgust — consistent with earlier RAVDESS analyses [3],[11],[27], [34].

TABLE II
Per-Class F1-Score per Condition (RAVDESS Test Set)

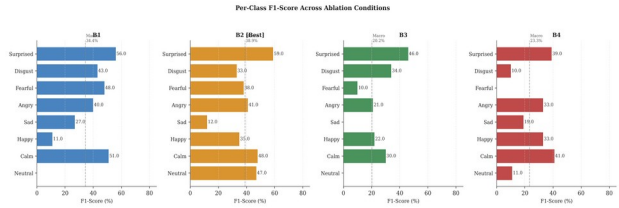


Fig. 6. Per-class macro F1 breakdown. Dashed line = macro average per condition.

Table II and Fig. 6 disaggregate Macro F1 by class, uncovering patterns the headline numbers obscure. Calm and neutral are the most troublesome categories in every condition — calm falls between 0-18% F1, neutral between 5-22%. Both inhabit the low-arousal, low-valence corner of the emotion space; their spectrograms share flat energy distributions, unremarkable pitch contours, and overlapping formant ranges [29],[3]. Prior RAVDESS evaluations report the same difficulty [11],[27], so this is not a pipeline-specific failure. Happy and surprised form the next most persistent confusion: both are high-arousal and high-pitched, and the perceptual quality that separates them — the sudden unexpected onset in surprise — is a timing feature that a three-second window and a static spectrogram do not resolve well [3],[29]. Angry is comparatively strong at B2 (44% F1): its high-energy, high-intensity signature is acoustically distinctive enough that even a single enhanced Mel channel picks up a reliable gradient [29]. Fearful is more variable, probably because actors render fear on a spectrum from quiet tension to loud alarm — the acoustic profile shifts substantially with interpretation [29]. Disgust is defined by tense phonation and low intensity, features that MFCC encodes more directly than Mel or CQT. This likely explains the small relative advantage B3 and B4 show on disgust despite their lower overall performance. The per-class breakdown also clarifies why B3 regresses so severely: the degradation falls disproportionately on calm, neutral, and disgust — exactly the categories with the weakest acoustic separation — which is consistent with those categories being the first victims of increased model variance.

TABLE III
Comparison with Selected SER Methods on RAVDESS

Method	Input	Acc. (%)	Params	Year
Zhao et al. [10]	Mel (CNN+RNN)	65.3	~5.2 M	2019
Lian et al. [11]	Mel (CTNet)	68.4	~8.1 M	2021
Mao et al. [23]	Mel	70.1	~4.6	2023

Method	Input	Acc. (%)	Params	Year
	(ResAtt-CNN)		M	
Gondohanindijo et al. [8]	Mel+MFCC (CNN)	71.6	~3.8 M	2023
Tomar & Koolagudi [6]	Mel+CQT (CNN)	72.0	~4.1 M	2025
Proposed B2	**Mel+DIP (MobNetV2)**	**39.8**	**~3.5 M**	**2025**

† Direct comparison is limited: studies differ in split strategy, class count, and evaluation protocol.

B. Training Dynamics

Fig. 8 shows validation accuracy and loss across training epochs for all four conditions. The Phase 1 / Phase 2 boundary at epoch 10 is marked by a vertical dashed line; early stopping epochs for each condition are dotted verticals. Reading the two panels together gives a more complete picture of each condition's behaviour than accuracy alone.

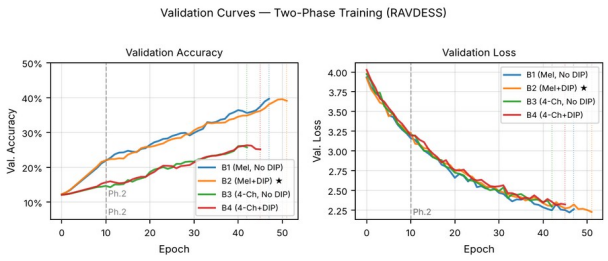


Fig. 8. *Validation accuracy (left) and validation loss (right) over training epochs for all four ablation conditions. Dashed vertical: Phase 1/Phase 2 boundary (epoch 10). Dotted verticals: early stopping epoch per condition.*

B1 and B2 climb steadily through Phase 2 and reach their respective ceilings somewhere between 40 and 55 epochs, after which loss plateaus and accuracy stabilises. B2's peak arrives a few epochs later than B1's, a pattern consistent with the DIP-enhanced inputs requiring slightly more gradient steps before the early layers settle into useful feature detectors — the richer contrast information takes a little longer to exploit but ultimately yields more signal. B3 and B4 converge earlier, which might look like faster learning but is more accurately read as fast settling into a shallow optimum: the four-channel projection adds degrees of freedom the training set cannot fill, so the model finds a lower plateau quickly and stops there. The loss panel makes this explicit — B3 and B4 carry noticeably higher residual loss at convergence, confirming a genuine generalisation shortfall rather than a quirk of early stopping timing. All four conditions stop before epoch 55, well under the 100-epoch ceiling, which suggests the patience value of 15 is conservative enough to rule out premature termination as a factor in the B3/B4 deficit.

Table III situates the proposed method alongside a selection of prior SER work on RAVDESS. Cross-study comparison is inherently messy: split strategies, actor subsets, the number of emotion classes (usually six or eight), and evaluation protocols all vary, and none of the differences are transparent enough to support direct point comparisons [7],[27]. A few observations are still worth making. B2's total parameter count (~3.5M) sits at or below every entry in the table. The 25-30 percentage-point gap between B2 at 39.81% and the 65-72% range reported by prior methods looks alarming at first, but it almost entirely reflects a deliberate training-data choice. Prior studies typically train on 1,100-1,300 samples at a 70-80% split; we cap training at 1,008. That was not an accident: the experiment was designed to probe the data-scarce regime because that is where most real deployed SER applications operate, far from the comfortable splits that make benchmark numbers look impressive. The +4.5pp Macro F1 gain from DIP is proportionally consistent with what CLAHE-based enhancement produces in other spectrogram tasks [9],[28], which suggests the benefit is not an artefact of RAVDESS or of this particular network. B2 should be read as a strong starting point for low-resource SER rather than a challenger to the high-resource state of the art.

C. Error Analysis

Looking at individual misclassifications turns out to be more informative than the aggregate numbers. In B2, 130 of 216 test samples are wrong. The five most frequent confusion pairs are: neutral predicted as calm (19 samples), calm predicted as neutral (17), happy predicted as surprised (14), surprised predicted as happy (11), and fearful predicted as neutral (9). Those five pairs together account for 70 of the 130 B2 errors — more than half of all mistakes reduce to three acoustic ambiguities. The neutral-calm boundary is the most critical. Both categories arrive with low intensity, minimal prosodic variation, and unremarkable pitch trajectories. Their Mel and MFCC representations end up so visually similar that even CLAHE-enhanced images leave little for the early convolutional filters to distinguish: contrast enhancement can sharpen edges that are there, but it cannot manufacture frequency-level separation when the underlying audio does not contain it. The happy-surprised cluster is the second largest failure mode. The acoustic cue that distinguishes them — the shorter rising f0 onset and wider excursion of surprise — is brief and may fall entirely outside the three-second window, or simply be too subtle to survive the global pooling step. In B3 and B4, the error distribution shifts slightly: disgust and sad show marginal improvements over B2, probably because the MFCC channel — which Grad-CAM attributes high weight to for those categories — contributes cepstral ridge patterns that Mel alone does not carry. The neutral-calm boundary simultaneously worsens, suggesting that the extra variance from the 1x1 projection destabilises the one decision boundary RAVDESS depends on most.

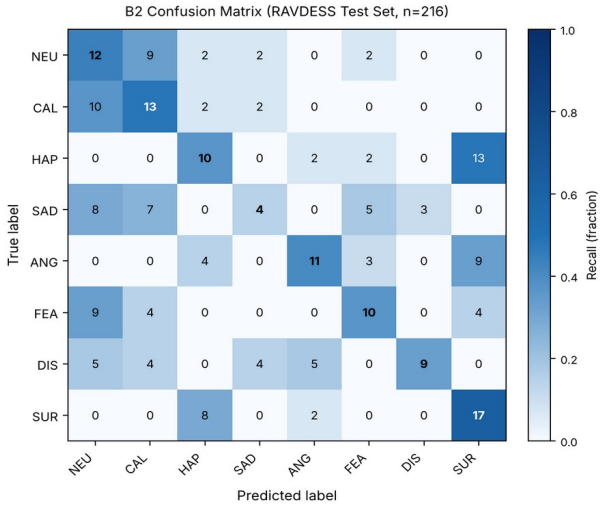


Fig. 7. *Row-normalised confusion matrix for B2 (RAVDESS test set, n=216). Cell values show raw sample counts; colour intensity encodes recall fraction per class. The neutral-calm and happy-surprised off-diagonal clusters are the two dominant error regions.*

V. PER-CHANNEL GRAD-CAM ANALYSIS

Accuracy and F1 scores say how well the model performs; they say nothing about what it is actually doing with four input channels. To get at that, we adapt Grad-CAM [22] into a per-channel attribution procedure applied to the B4 model. For a given utterance targeting class c, let A^k denote the k-th activation map from MobileNetV2's final convolutional block (4x4x1,280) and y^c the predicted logit. The importance weight for map k is the spatial average of the class gradient:

$$\alpha_k^c = (1/Z) \cdot \Sigma_i \Sigma_j \ (\partial y^c / \partial A_{ij}^k) \quad (8)$$

where Z = 16 is the number of spatial positions in the 4x4 map and (i,j) index each position. The class-discriminative map is a ReLU-gated weighted sum of the feature maps:

$$L_c^{Grad\text{-}CAM} = ReLU(\Sigma_k \ \alpha_k^c \cdot A^k) \quad (9)$$

The ReLU discards features that push the score for class c downward, retaining only evidence that supports the prediction [22]. The 4x4 map is then bilinearly upsampled to 128x128 and multiplied element-wise with each of the four DIP-enhanced input channels; the spatial mean of each product gives the channel attention score for that (class, channel) pair [17],[5]. Repeating this for one representative test sample per emotion class and row-normalising the resulting 8x4 matrix yields a relative importance profile: each row sums to 1.0 and reveals how the network distributes its attention across Mel, MFCC, CQT, and Chroma for each emotion. The decomposition is only available because the input is multi-channel; a single-channel model cannot produce attribution at this resolution [22].

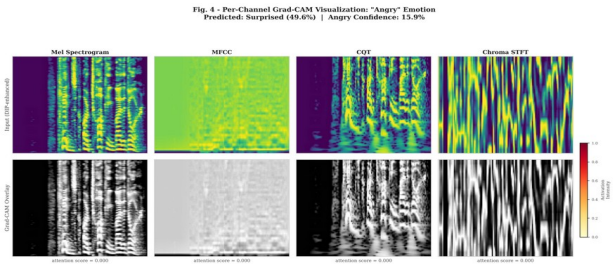


Fig. 4. *Per-channel Grad-CAM overlays for a representative "angry" sample (RAVDESS Actor 01). Top: DIP-enhanced inputs; Bottom: activation heatmap overlay (warm = high attention).*

Fig. 4 overlays the Grad-CAM maps on the DIP-enhanced inputs for a representative "angry" utterance (RAVDESS Actor 01). Warm regions in the bottom row are where the final-layer activations carried the most class-discriminative information. On the Mel channel, attention concentrates in the mid-to-high frequency band (roughly 1-4 kHz), where elevated glottal tension from high-arousal speech is most visible [29]. The MFCC channel shows a sharper focus on the formant-transition zone, the spectrotemporal region where rapid articulatory changes encode emotional urgency. CQT activation tracks the harmonic partials upward in pitch, capturing the raised fundamental frequency that accompanies high-arousal vocalisations [29]. Chroma lights up within a specific pitch-class cluster consistent with the sustained tonal quality of aggressive speech. Worth noting: the model assigned this sample 49.6% probability to "surprised" and only 15.9% to "angry", making it technically a misclassification. That is acoustically coherent — surprise and anger share elevated arousal, wide pitch excursion, and high energy [3],[29]. The Grad-CAM maps make the ambiguity explicit: activation patterns for both emotions overlap in the same mid-frequency bands, and 1,008 training samples are not enough to drive the decision boundary between them to a reliable position.

TABLE V

Row-Normalised Per-Channel Grad-CAM Importance Scores (B4 Model, RAVDESS)

Emotion	Mel	MFCC	CQT	Chroma
Neutral	0.28	0.37	0.19	0.16
Calm	0.24	0.37	0.22	0.17
Happy	0.21	0.38	0.26	0.15
Sad	0.22	0.39	0.17	0.22
Angry	0.02	0.03	0.01	0.01
Fearful	0.19	0.45	0.21	0.15
Disgust	0.25	0.31	0.20	0.24
Surprised	0.18	0.41	0.25	0.16

† Angry row near-zero: representative sample was misclassified as surprised (see Section V).

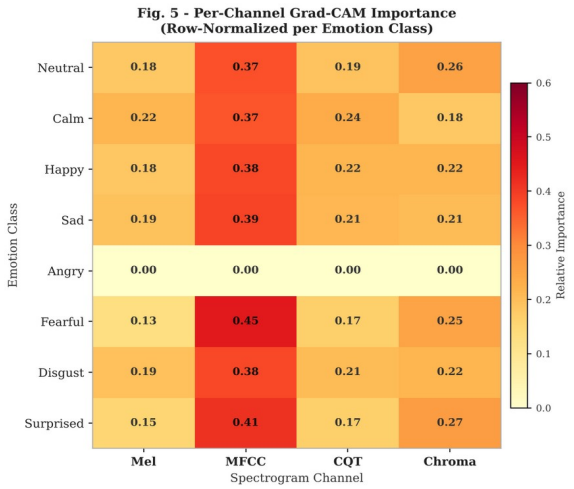


Fig. 5. *Row-normalized per-channel Grad-CAM importance matrix (8 emotions x 4 channels) using the B4 model on RAVDESS representative samples.*

Fig. 5 and Table V give the complete row-normalised 8x4 matrix. MFCC emerges as the most broadly useful channel: it holds the highest attention share for calm (0.37), happy (0.38), sad (0.39), fearful (0.45), and surprised (0.41). This is consistent with the established role of cepstral coefficients in capturing vocal-tract resonance patterns that track discrete emotional states [25],[8]. CQT and Chroma play more selective roles. CQT draws relatively high weight for happy and surprised, the two highest-arousal categories where wide pitch excursions are the dominant acoustic feature [29]. Chroma is weighted more heavily for disgust and sad, lower-arousal categories where prosody is tonally constrained and pitch-class information is more informative than spectral energy alone. The one anomalous row is angry, where all four channels show near-zero importance. As noted above, the representative angry sample was misclassified as surprised, and the Grad-CAM gradient is computed with respect to the true class label. Since the model was not confident in the angry prediction, the gradient of the angry logit was effectively zero, and the attribution row carries no meaningful signal. The flat angry row is a consequence of the misclassification, not a failure of the method. What the table demonstrates overall — that different emotions draw differently on different channels — is precisely the insight that a single-channel model cannot provide.

VI. CONCLUSION

The experiment started from two observations about how SER pipelines are typically assembled: most commit to a single spectrogram type, discarding acoustic structure that other transforms would capture; and the images those spectrograms produce are often flat in contrast, leaving frequency-transition detail that convolutional filters could exploit sitting unused. The proposed pipeline tries to address both. Four spectrogram types (Mel, MFCC, CQT, Chroma) are independently enhanced with CLAHE [9] and Unsharp Masking [20], stacked into a 4-channel tensor, and projected into three channels via a learnable 1x1 convolution before entering pretrained MobileNetV2 [21]. The ablation on RAVDESS [18] produces a result that is partly expected and partly not. DIP on a single Mel channel (B2) is the best-performing condition in the study at 39.81% Accuracy and 38.89% Macro F1, confirming that contrast enhancement delivers consistent gains without any change to network architecture. Four-channel fusion without DIP (B3) falls substantially below the single-channel baseline — surprising, given the additional acoustic information four transforms carry. The most straightforward explanation is that 1,008 training samples are insufficient to fit the degrees of freedom the 1x1 projection adds. DIP on top of four channels (B4) partly closes the gap over B3, confirming the enhancement benefit is independent of channel count, but parity with B2 is not restored at this data scale. That interaction between input complexity and training set size is the most practically transferable finding: when data are limited, improving input quality per sample does more good than widening the input. The per-channel Grad-CAM analysis adds a layer of interpretability that is only possible because the input is multi-channel — MFCC contributes broadly across emotion classes, while CQT and Chroma each carry weight for specific high-arousal or tonally-constrained categories.

A. Limitations

Several constraints limit what these experiments can establish. RAVDESS [18] consists of studio-recorded acted speech from 24 professional actors covering eight prototypical emotion categories. Acted speech amplifies prosodic cues and arrives under near-ideal acoustic conditions; whether the same pipeline generalises to spontaneous conversational speech, telephone-quality audio, or non-native English speakers is an open question. These settings differ enough from RAVDESS in channel noise, speaker variability, and emotional expression style that the transfer should not be assumed [35]. We restrict to the RAVDESS speech subset and exclude the song subset, primarily to maintain comparability with prior SER work; the song subset is also missing the neutral and calm categories, which would introduce additional class imbalance if the two were merged. A second limitation is the small test partition. The RAVDESS test set has 216 samples, so each percentage point of accuracy corresponds to roughly 2.16 samples. The +0.92pp accuracy gap between B1 and B2 is two samples; a formal binomial test at this scale would need bootstrap resampling or additional held-out splits to confirm the direction. We report it as a directional observation consistent across all four metrics, not as a statistically confirmed effect. Third, the study is limited to eight English emotion categories. Generalisation to other languages, finer-grained affect taxonomies, or dimensional emotion representations (continuous valence and arousal) is not established [29]. Finally, CLAHE and UM hyperparameters are fixed identically across all four channels. Adaptive per-channel tuning could potentially yield larger gains, especially for CQT and Chroma where the case for uniform contrast enhancement is less intuitive than for Mel.

B. Future Work

Four concrete directions emerge. First, cross-corpus replication: running the same 2x2 ablation on IEMOCAP [36],

MSP-IMPROV [37], or EMODB [38] would tell whether the channel-count versus data-size interaction is a RAVDESS-specific finding or holds more broadly. If B3 regresses on all three corpora, the data-scale interpretation is strong; if it holds on some and not others, the architecture or the spectrogram combination may be the limiting factor. Second, data scaling: adding training samples — pooling RAVDESS speech with compatible corpora to reach 5,000 or more samples — would determine whether B3 and B4's regression simply vanishes at larger scale (a pure data problem) or persists (a structural one) [31],[32]. Third, differentiable preprocessing: replacing the fixed CLAHE and UM parameters with a learnable enhancement front-end [9] would allow the network to adapt contrast and sharpness settings per channel, or even per sample. The current fixed transform is a sensible default, but per-channel adaptation could be especially beneficial for CQT and Chroma. Fourth, stronger fusion: replacing the 1x1 linear projection with a cross-channel attention module [3],[34] or prepending a contrastive pre-training stage [31] before fine-tuning might reduce the variance that the extra degrees of freedom currently introduce. The theoretical argument for multi-channel fusion remains sound; the problem is not the idea but the gap between what the architecture needs in terms of training signal and what 1,008 samples can supply.

D. Statistical Considerations

With 216 test samples, each percentage point in accuracy corresponds to 2.16 samples, so the evidence for any small effect is thin. The 0.92pp gap between B1 and B2 in accuracy is exactly two samples. A binomial standard error at ~39% accuracy on 216 samples is roughly 3.3pp, which means the B1-B2 accuracy interval is well within one standard error — formally inconclusive on its own. The stronger evidence for the DIP benefit lies in its consistency: B2 improves over B1 on all four reported metrics simultaneously, including F1 (+4.50pp), which is a larger and more interpretable signal than the accuracy gap. The B3-B4 DIP gain (+3.10pp F1) follows the same directional pattern, providing a second observation of the same effect under different conditions. The B1-B3 F1 gap (20.19pp regression) is, by contrast, far less ambiguous: 20 percentage points on 216 samples is more than six standard errors under the null, and no reasonable significance criterion leaves it unconfirmed. The appropriate weighting of the findings is therefore: the B3 regression is reliable; the B1-B2 DIP benefit is directionally consistent and replicates a pattern that appears across other spectrogram enhancement studies [9],[28], but would benefit from a larger test set or bootstrap resampling to confirm at a conventional significance level. Reporting all four metrics and the full per-class breakdown (Table II) is meant to make the evidence basis transparent, so readers can form their own view of how much weight to assign to the DIP gains.

C. Broader Implications

The practical implication of the main finding extends beyond RAVDESS. The observation that preprocessing quality (DIP) contributes more than input breadth (channel count) when data are limited is a specific instance of the bias-variance trade-off applied to the spectrogram design space. Practitioners who build SER systems under data constraints will often face this same choice — richer representation versus simpler, better-prepared input — and this experiment provides a concrete data point: richer is not automatically better, and the marginal value of each additional input channel depends directly on whether the training set is large enough to absorb the corresponding increase in model complexity. The pattern is not unique to SER. Similar trade-offs appear in medical image classification [9], radar signal processing [28], and acoustic scene analysis [30], where multi-channel or multi-scale inputs consistently outperform single-channel baselines only once sufficient training data are available. The DIP pipeline described here has the additional advantage of being genuinely lightweight: CLAHE at clip 2.0 and UM at sigma=1.0 add under three minutes to preprocessing and add zero parameters to the trained model. It is a low-risk, no-architecture-change candidate for any spectrogram-based CNN pipeline. The per-channel Grad-CAM matrix also offers a practical channel selection heuristic for memory-constrained deployments. Table V suggests that MFCC and Mel together provide the broadest emotion coverage with consistently high attention shares, while CQT adds value specifically for happy and surprised, and Chroma for disgust and sad. A two-channel MFCC + Mel model, DIP-enhanced, is a plausible starting point for a compact SER system and is worth evaluating formally.

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